Report 13/04/2026

Perception Learning (V-JEPA Pretraining)

- Use a ViT encoder with an EMA target encoder to learn spatiotemporal representations from front-camera video.
- Representation objective (instead of pixel reconstruction):

$$\min_{\theta,\phi} \|P_\phi(\Delta y, E_\theta(x)) - \mathrm{sg}(E_{\bar{\theta}}(y))\|_2$$

- Academic motivation: reduce representation collapse and improve semantic invariance for downstream planning.
- VinFast setup: short clips, front-camera focused, self-supervised training without 3D perception labels.

Multimodal Trajectory Distillation (MTD)

- Build a trajectory vocabulary with **8192 anchors** via K-means over human driving trajectories.
- Replay anchors in simulator and score with **EPDM/EPDMS**; keep high-score trajectories as **pseudo-teachers**.
- A *min-over-N* style objective encourages proposal coverage of multiple safe modes instead of single-mode collapse.
- Practical effect: improved trajectory diversity in uncertain traffic scenarios.

Planner and Momentum-aware Selection

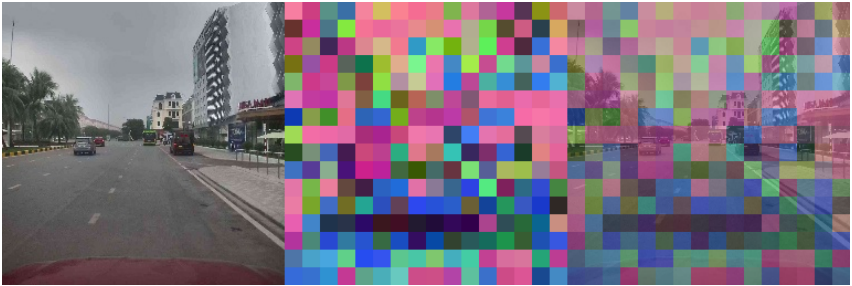
- The planner takes perception features + ego status and generates multiple trajectory proposals (WADA refinement).
- A scorer ranks proposals using safety, progress, and dynamic feasibility signals.
- Momentum-aware update enforces temporal consistency:

$$S \leftarrow \frac{7S + S_c}{8}$$

- Target behavior: safe and smooth trajectories with reduced frame-to-frame switching.

Current Implementation Results

- Built the VF → V-JEPA2 dataset pipeline (frame-to-clip conversion + train/val index files).
- Completed tiny/smoke pretraining end-to-end with decreasing loss and valid outputs/checkpoints.
- Added PCA feature visualization (frame overlay + latent PCA map) for representation-level diagnostics.



Requirements, and Next Plans

- **VinFast requirements:** PDMS-focused validation, and trajectory visualization on VF data.
- **Key gaps for PDMS on raw VF data:**
  - Missing 3D agent boxes/tracks and velocities $\Rightarrow$ No Collision/Time-to-Collision cannot be evaluated reliably.
  - Missing HD map and route annotations $\Rightarrow$ Drivable Area/Ego Progress cannot be computed on VF scenes.
  - Only comfort-related signals are partially available from NAV/vehicle CSV.
  - NavSim map API is tied to nuPlan map locations, so VF scenes fail at direct map loading.
- **Next week plan:**
  - *Trajectory Analysis:* explore methods for building and enriching a trajectory dataset from raw VF data.
  - *Planner environment:* investigate alternative simulation environments (e.g., CARLA, nuPlan replay, or synthetic benchmarks) that can replace NavSim when full VF sensor/map data is unavailable.